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Chapter 2

Introduction to TCP/IP

You have to be an expert at the TCP/IP protocol in order to pass the Cisco ICND1 and ICND2 exams. In order to prepare you for TCP/IP-related questions, this chapter reviews critical points such as IP addressing, protocols, and common ports. The next chapter builds off this and covers more advanced TCP/IP questions, but let's hit the basics first!

The Questions You'll Work On

In this chapter, you'll review questions concerning the following topics:

- Understanding IP addressing topics, such as the default address classes
- Recognizing the different protocols that make up the TCP/IP protocol suite, such as TCP, UDP, IP, ARP, and ICMP
- Demonstrating on the new ICND1 exam your knowledge of IPv6 and the different address types
- Knowing the common TCP and UDP ports that are used by applications
- Converting a binary value to a decimal value and vice versa

What to Watch Out For

Don't let common mistakes trip you up; watch for the following when working with these questions:

- There are three core address classes, known as class A, class B, and class C. Class A addresses have the first octet with a value that ranges from 1 to 127. Class B addresses have the first octet with a value that ranges from 128 to 191. Class C addresses have the first octet with a value that ranges from 192 to 223.
- Watch for the use of common protocols such as IP for logical addressing and routing. ICMP does error and status reporting, TCP is responsible for reliable delivery, UDP is responsible for unreliable delivery, and ARP is responsible for converting the logical address to the physical address.
- Know the port values used by common applications such HTTP (80), Telnet (23), SSH (22), HTTPS (443), FTP (21 and 20), SMTP (25), POP3 (110), and DNS (53).
- Know the IPv6 benefits and types of addresses. Link-local addresses are similar to APIPA addresses and start with FE80. Unique local addresses are similar to private IP addresses and start with FC00, and finally, global addresses are like public IP addresses and start with 2000. The IPv6 loopback address is ::1.

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IP Addressing

39–54 Choose the best answer(s).

39. Which of the following are considered class A addresses? (Select all that apply.)

(A) 129.45.10.15

(B) 10.35.87.5

(C) 131.15.10.12

(D) 192.156.8.34

(E) 121.59.87.32

(F) 210.45.10.112

40. Sue is reviewing IP addressing basics and asks you which of the following is considered a class A private address.

(A) 24.56.10.12

(B) 192.168.0.5

(C) 172.16.45.10

(D) 10.55.67.99

41. What is the default subnet mask of a system with the IP address of 189.34.5.67?

(A) 255.0.0.0

(B) 255.255.0.0

(C) 255.255.255.0

(D) 255.255.255.255

42. What is the default subnet mask of a class C address?

(A) 255.0.0.0

(B) 255.255.0.0

(C) 255.255.255.0

(D) 255.255.255.255

43. Which of the following is a class B private address?

(A) 24.56.10.12

(B) 192.168.0.5

(C) 172.16.45.10

(D) 10.55.67.99

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44. Are the systems of 201.45.3.56 and 201.45.5.20 on the same network?

(A) Yes

(B) No

45. Which of the following addresses are considered invalid addresses to assign to a host on the network? (Select all that apply.)

(A) 10.254.255.255

(B) 120.127.34.10

(C) 190.34.255.255

(D) 136.45.68.22

(E) 202.45.6.0

(F) 127.87.3.22

46. The decimal number of 137 is which of the following in binary?

(A) 10001001

(B) 10101001

(C) 11001001

(D) 10000101

47. Are the systems with the IP addresses of 121.56.78.10 and 121.45.6.88 on the same network?

(A) Yes

(B) No

48. What is the default subnet mask of 130.56.78.10?

(A) 255.0.0.0

(B) 255.255.0.0

(C) 255.255.255.0

(D) 255.255.255.255

49. Your system has an IP address of 131.107.45.10. Which of the following systems is on the same network as you?

(A) 131.56.10.12

(B) 130.107.45.10

(C) 131.107.22.15

(D) 113.107.45.11

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50. Which of the following addresses are considered invalid addresses to assign to a host on the network? (Select all that apply.)

(A) 216.83.11.255

(B) 12.34.0.0

(C) 10.34.15.22

(D) 131.107.0.0

(E) 127.15.34.10

(F) 189.56.78.10

51. The binary number of 01101101 is which of the following decimal values?

(A) 101

(B) 109

(C) 135

(D) 143

52. Which class address always has the value of the first bits in the IP address set to 110?

(A) Class A

(B) Class B

(C) Class C

(D) Class D

53. Which class address always has the first two bits of the IP address set to 10?

(A) Class A

(B) Class B

(C) Class C

(D) Class D

54. Which IP address class always has the first bit in the address set to 0?

(A) Class A

(B) Class B

(C) Class C

(D) Class D

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TCP/IP Protocols

55–71 Choose the best answer(s).

55. Which protocol is responsible for converting the logical address to a physical address?

(A) TCP
(B) IP
(C) ICMP
(D) ARP

56. The ARP request is sent to which of the following layer-2 destination addresses?

(A) FF-FF-FF-FF-FF-FF
(B) 192.168.0.255
(C) 00-FF-00-FF-00-FF
(D) 00-00-00-00-00-00

57. What are the three phases of the TCP three-way handshake?

(A) ACK/SYN, SYN, ACK
(B) SYN, ACK/SYN, ACK
(C) ACK/SYN, ACK, SYN
(D) SYN, ACK, SYN/ACK

58. Which of the following does TCP use to guarantee delivery?

(A) Source and destination IP address
(B) Source and destination port
(C) Sequence numbers and acknowledgements
(D) Sequence numbers and ports

59. What TCP/IP protocol is responsible for logical addressing and routing functions?

(A) TCP
(B) IP
(C) ICMP
(D) UDP

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60. Which transport layer protocol is responsible for unreliable delivery?

(A) TCP
(B) IP
(C) ICMP
(D) UDP

61. Which TCP/IP protocol is responsible for error and status reporting?

(A) TCP
(B) IP
(C) ICMP
(D) UDP

62. The router looks at which field in the IP header to decide where to send the packet?

(A) Source IP address
(B) Destination IP address
(C) Source MAC address
(D) Destination MAC address

63. What flags are set on the second phase of the three-way handshake?

(A) FIN/SYN
(B) ACK/FIN
(C) ACK/SYN
(D) ACK/RST

64. Which flag is set in a TCP packet that indicates a previous packet was received?

(A) PSH
(B) FIN
(C) ACK
(D) RST

65. You wish to allow echo request messages to pass through the firewall. What ICMP type is used in an echo request message?

(A) 0
(B) 8
(C) 3
(D) 11

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66. Which of the following are fields found in the IP header? (Select all that apply.)

(A) Sequence number

(B) Destination port

(C) Source IP address

(D) Type

(E) Time to Live

(F) SYN flag

67. Which of the following are considered application layer protocols?

(A) TCP

(B) IP

(C) FTP

(D) ICMP

68. Which transport layer protocol is responsible for reliable delivery?

(A) UDP

(B) ICMP

(C) IP

(D) TCP

69. What ICMP type is used by the echo reply message?

(A) 0

(B) 3

(C) 5

(D) 8

70. The security administrator for your network has asked that you block ping messages from entering your network. What protocol would you block?

(A) ICMP

(B) IP

(C) TCP

(D) UDP

71. Which TCP flag is responsible for dropping a connection at any point in time?

(A) PSH

(B) FIN

(C) ACK

(D) RST

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IPv6

72–78 Choose the best answer(s).

72. Which of the following is the IPv6 equivalent to 127.0.0.1?

(A) ::127

(B) 127::1

(C) ::1

(D) FE80::

73. Your manager has been hearing a lot about IPv6 addressing, and asks you which of the following statements are true about IPv6 unicast addresses? (Select two.)

(A) A link-local address starts with FE00.

(B) A global address starts with 2000.

(C) The loopback address is 127::1.

(D) When an interface is assigned a global address, it is allowed only one IP6 address.

(E) The loopback address is ::1.

74. Which of the following is an example of an IPv6 link-local address?

(A) ff00::f407:622c:a0ce:90cc

(B) fe80::f407:622c:a0ce:90cc

(C) fe08::f407:622c:a0ce:90cc

(D) 2001::f407:622c:a0ce:90cc

75. Your manager has asked you what some of the benefits of transitioning from IPv4 to IPv6 are. (Select two.)

(A) IPSec is optional

(B) No broadcast messages

(C) A 64-bit address scheme

(D) Telnet passwords are encrypted

(E) Automatic configuration

76. Your manager is evaluating IPv6 transitioning technologies and is wondering which IPv6 tunneling method encapsulates the IPv6 data into an IPv4 user datagram to travel over the Internet and can pass through NAT devices.

(A) 6-to-4

(B) Dynamic IP

(C) Teredo

(D) Dual stack

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77. You are evaluation how you are going to slowly transition to IPv6. What three techniques can be used to allow migrating from IPv4 to IPv6? (Choose three.)

(A) Broadcast addressing

(B) Translation between IPv6 and IPv4

(C) Multicast to unicast translation

(D) Use of dual-stack routing

(E) Use tunneling protocols

78. What IPv6 address type is similar to an IPv4 private IP address and starts with FC00?

(A) Global address

(B) Loopback address

(C) Link-local address

(D) Unique local address

TCP/IP Ports

79–83 Choose the best answer(s).

79. SMTP uses which of the following ports?

(A) UDP 53

(B) TCP 21

(C) TCP 23

(D) TCP 25

80. FTP uses which of the following ports?

(A) 25

(B) 23

(C) 21

(D) 53

81. Which of the following represents the port used by POP3?

(A) 21

(B) 110

(C) 143

(D) 443

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82. Which of the following are true statements about TCP ports? (Select all that apply.)

(A) FTP uses TCP ports 21 and 20.

(B) FTP uses TCP ports 11 and 12.

(C) HTTP uses TCP port 53.

(D) HTTP uses TCP port 80.

(E) SMTP uses TCP port 53.

(F) SMTP uses TCP port 25.

83. Which of the following are true statements about ports? (Select all that apply.)

(A) DHCP uses UDP 67/68.

(B) DHCP uses TCP 67/68.

(C) SNMP uses UDP 161.

(D) DNS uses UDP 25.

(E) SNMP uses TCP 161.

(F) DNS uses UDP 53.

Troubleshooting

84–85 Choose the best answer(s).

84. Sue is unable to surf the Internet and calls you over to her desk to troubleshoot. You use ipconfig on the system and notice the following IP configuration. What is the problem?

```
Local Area Network Connection:
IPv4 Address…. … .: 169.254.34.56
Subnet Mask… …. .: 255.255.0.0
Default Gateway… …:
```

(A) The DNS server is down.

(B) The subnet mask is incorrect.

(C) The router is down.

(D) The DHCP server is down.

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85. Bob is unable to surf the Internet and calls you over to his desk to troubleshoot. You use ipconfig on the system and notice the following IP configuration. What should you do?

```
Local Area Network Connection:
IPv4 Address…. … .: 192.19.210.79
Subnet Mask… …. .: 255.255.255.0
Default Gateway… …: 192.19.0.1
```

(A) Change his IP address to 169.254.210.79.

(B) Change his IP address to 192.19.0.79.

(C) Change his subnet mask to 255.0.0.0.

(D) Remove his default gateway setting.